

Semiclassical Tunneling Calculations

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We show that a semiclassical approximation for calculating thermally averaged transition-state-theory transmission coefficients for realistic effective potential energy barriers for a separable reaction coordinate generally provides accuracy of 10% or better compared to accurate quantum mechanics for the same potential energy barrier. This error is smaller than the error generally caused by the separable approximation itself. We show how to perform the semiclassical calculations conveniently and efficiently.

1. Introduction

Tunneling is important in many chemical reactions. In transition-state theory, the reaction coordinate is assumed separable from the other degrees of freedom, and tunneling is included by multiplying the rate constant computed with the usual classical treatment of reaction-coordinate motion by a "tunneling correction", $\kappa(T)$, which may be considered to be part of the transmission coefficient.¹⁻⁴ Let the effective potential energy barrier to motion along the reaction coordinate s be $V(s)$ with a maximum, $V_{\max}$, at $s = s_{\max}$. Then the quantum correction on the reaction-coordinate motion for temperature T is conventionally defined as the ratio of the thermally averaged quantum mechanical transmission probability, $T(E)$, to the thermally averaged classical transmission probability for the same potential energy barrier, i.e.

$$\kappa(T) = \frac{\int_0^\infty T(E) \exp(-\beta E) dE}{\int_{V_{\max}}^\infty \exp(-\beta E) dE} \quad (1)$$

$$= \beta \exp(\beta V_{\max}) \int_0^\infty T(E) \exp(-\beta E) dE \quad (2)$$

where β is $(kT)^{-1}$ and k is Boltzmann's constant. Notice that the integral extends up to infinity because the quantum correction on the reaction-coordinate motion must account for nonclassical reflection at energies above the barrier height as well as nonclassical transmission at energies below the barrier height. For this reason "tunneling correction" is a misnomer, although the tunneling region does dominate the thermal average under the typical conditions where the quantal correction has been employed. In many cases it is sufficient to use simple approximations to $\kappa(T)$, for example, approximations based on parabolic or Eckart-type approximations to $V(s)$, for which $T(E)$ can be computed analytically.^{1,5-11} For more quantitative work or for cases where the parabolic or Eckart-type approximations do not provide an adequate fit to the shape of the effective potential energy barrier, one can calculate $T(E)$ numerically and numerically integrate eq 2 to obtain the tunneling correction.^{12,13} Although such a quantum mechanical calculation is straightforward for any one-dimensional barrier, it requires more care and effort than is warranted in many cases. Thus it is useful to have a simpler and reasonably accurate method for calculating the thermally averaged quantal correction on reaction-coordinate motion which can be applied for any reasonably smooth barrier shape. Such a method may be based on a semiclassical approximation to the transmission coefficient.¹⁴⁻¹⁹ The purpose of the

present work is to present such a semiclassical approximation for the tunneling correction in a computationally convenient form and to test its accuracy against complete quantum mechanics. The method we propose is very economical and has accuracy sufficient for many problems.

The present test of the accuracy of semiclassical mechanics for barrier penetration also has implications for the accuracy attainable by more sophisticated semiclassical methods such as classical S matrix theory²⁰⁻²⁴ or semiclassical approximations to nonseparable transition-state theory.²⁵ The present method may also be used, with appropriate modifications, for including quantum mechanical effects on reaction-coordinate motion in variational transition-state theory. This will be discussed in a separate article.

2. Theory

The semiclassical approximation to the transmission probability for an energy E below the barrier height $V_{\max}$ is given by¹⁴⁻¹⁹

$$T^S(E) = \{1 + \exp[2\theta(E)]\}^{-1} \quad (3)$$

where the barrier penetration integral is given by

$$\theta(E) = \hbar^{-1} \int_{s_-}^{s_+} ds \{2\mu[V(s) - E]\}^{1/2} \quad E < V_{\max} \quad (4)$$

s_+ and s_- are the classical turning points, i.e., the locations at which $V(s) = E$, and μ is the reduced mass for the motion along the reaction coordinate. The threshold energy is

$$E_0 = \max[V(s = -\infty), V(s = +\infty)] \quad (5)$$

At $E = E_0$, the semiclassical transmission probability does not tend to zero. Rather it tends to a finite value, although this value is usually very small for systems for which tunneling corrections are important. As the energy approaches the barrier height $V_{\max}$ the barrier penetration integral tends to zero and $T^S(E)$ tends to the value 1/2. In order to perform a Boltzmann average as indicated in eq 2, it is necessary to approximate the transmission probability for energies above the barrier height. We accomplish this as follows. First, approximate the potential near the maximum of the barrier through quadratic terms in a Taylor series

$$V(s) \simeq V_{\max} - \frac{1}{2}F_s(s - s_{\max})^2 \quad (6)$$

where

$$F_s = -\left. \frac{\partial^2 V}{\partial s^2} \right|_{s=s_{\max}} \quad (7)$$

The barrier penetration integral (4) for energies near but

below the barrier height can be approximated by

$$\theta(E) \simeq \pi \hbar^{-1} [\mu/F_s]^{1/2} (V_{\max} - E) \quad (8)$$

For energies near and above the barrier height the parabolic approximation to the barrier penetration integral can also be obtained.^{14,16} The turning points are purely imaginary for such energies, however, $\theta(E)$ is again given by eq 8. Therefore, for energies near the barrier height

$$\theta(V_{\max} + \epsilon) \simeq -\theta(V_{\max} - \epsilon) \quad \epsilon \simeq 0 \quad (9)$$

Equation 3 then yields

$$T^S(V_{\max} + \epsilon) \simeq 1 - T^S(V_{\max} - \epsilon) \quad \epsilon \simeq 0 \quad (10)$$

Because of the Boltzmann factor in eq 2, high energies make only a small contribution to the thermally averaged transmission coefficient. Thus we use eq 9 for all energies greater than the barrier height. Combining these results, our semiclassical prescription for the transmission probabilities is given by

$$\begin{aligned} T^S(E) &= \{1 + \exp[2\theta(E)]\}^{-1} & E_0 \leq E \leq V_{\max} \\ &= 1 - T^S(2V_{\max} - E) & V_{\max} \leq E \leq 2V_{\max} - E_0 \\ &= 1 & 2V_{\max} - E_0 < E \end{aligned} \quad (11)$$

The semiclassical approximation $\kappa^S(T)$ to the transmission coefficient can be obtained by substituting eq 11 into eq 2 which gives

$$\kappa^S(T) = 1 + 2\beta \int_{E_0}^{V_{\max}} dE \sinh[\beta(V_{\max} - E)] T^S(E) \quad (12)$$

To derive eq 12, we made a change of variables motivated by a related derivation of Shavitt.⁷ Thus the calculation of $\kappa^S(T)$ requires performing numerical quadratures of eq 4 and 12.

3. Computational Details

To reduce the barrier penetration integral (4) to the standard interval $(-1,1)$, we make the change of variables

$$s = A(1 + x) \quad (13)$$

where

$$A = (s_+ - s_-)/2 \quad (14)$$

This yields

$$\theta(E) = [(2\mu)^{1/2}A/\hbar] \int_{-1}^1 dx [V(x) - E]^{1/2} \quad (15)$$

Equation 15 could now be integrated by a Gauss-Legendre quadrature, but it is more efficient to proceed as follows. Consider, for motivation, a parabolic approximation to $V(x)$:

$$V(x) \simeq V_{\max} - (V_{\max} - E)x^2 \quad (16)$$

This yields

$$\theta(E) \simeq [2\mu(V_{\max} - E)]^{1/2}(A/\hbar) \int_{-1}^1 dx (1 - x^2)^{1/2} \quad (17)$$

and suggests the change of variables

$$dg = (4/\pi)(1 - x^2)^{1/2} dx \quad (18)$$

so that

$$g(x) = (2/\pi)[x(1 - x^2)^{1/2} + \arcsin x] \quad (19)$$

The proportionality constant in eq 18 and the constant of integration in eq 19 are determined by the requirements

$g(1) = 1$ and $g(0) = 0$. Making this change of variables in eq 15 yields

$$\theta(E) = [(2\mu)^{1/2}\pi A/4\hbar] \int_{-1}^1 dg \left\{ \frac{V[s(g)] - E}{1 - x(g)^2} \right\}^{1/2} \quad (20)$$

If $V(s)$ is actually a parabola between the classical turning points, then the integrand of eq 20 is a constant. In general, numerical integration of eq 20 converges faster than numerical integration of eq 15.

To integrate eq 20 we must find $x(g)$ by inverting eq 19 (an algorithm for this is given below) and $s(g)$ by combining this result with eq 13. Let the weights and abscissas for an N -point quadrature scheme on the interval $(-1,1)$ be w_i and g_i , respectively, and let

$$x_i = x(g_i) \quad (21)$$

Applying the quadrature scheme to eq 20, we then obtain

$$\theta(E) = [(2\mu)^{1/2}\pi A/4\hbar] \sum_{i=1}^N W_i \{V[A(1 + x_i)] - E\}^{1/2} \quad (22)$$

where

$$W_i = (1 - x_i)^{-1/2} w_i \quad (23)$$

Equation 22 is the form recommended for computation.

In order to use the quadrature scheme (22) one must invert eq 19 numerically. Although this is not difficult, a further simplification is achieved by noticing that, once a set of x_i is computed for a given quadrature scheme, it can be reused for calculations with other energies and other potential energy functions. The actual numerical inversion can be accomplished by Newton's method of iteration. This requires the derivative which is given in eq 18 and yields the following equation for the $(i + 1)$ st iterate in terms of the i th iterate:

$$x^{(i+1)} = \frac{x^{(i)}}{2} - \frac{\arcsin x^{(i)}}{2[1 - (x^{(i)})^2]^{1/2}} + \frac{\pi g}{4[1 - (x^{(i)})^2]^{1/2}} \quad (24)$$

We encountered no convergence difficulties.

We found it convenient to use Kronrod's method of quadrature.²⁶ In this method one performs an n -point Gauss-Legendre quadrature and a $(2n + 1)$ -point improved quadrature where n of the abscissae of the latter are constrained to be the same as in the former. Thus for the cost of $(2n + 1)$ evaluations of the potential energy function one can evaluate both a $(2n + 1)$ -point estimate of the integral and an n -point Gauss-Legendre estimate. For several values of n we have evaluated the $(2n + 1)$ values of x_i and W_i for the improved quadrature and the n values of W_i for the Gauss-Legendre quadrature. The results are given as part of a Fortran computer program in Appendix 2 (see paragraph at end of text regarding supplementary material). We used values of $(2n + 1)$ in the range 9-19 and found that they were sufficient to converge $\kappa^S(T)$ to 1% or better even at 200 K, where convergence is slower than at higher temperatures.

Some potential functions have more than one local maximum. For such cases, at some energies, there exists more than one pair of classical turning points and we perform a $(2n + 1)$ -point quadrature of $\theta(E)$ for each classically forbidden region. Then the overall barrier penetration integral is the sum of these quantities.

Having obtained $\theta(E)$ we calculate $T^S(E)$ by eq 11 and integrate eq 12 numerically. The numerical integration of eq 12 was also accomplished by Kronrod's method, using a $(2m + 1)$ -point improved quadrature. Improved quadratures with 9-15 points were found to be sufficient to calculate $\kappa^S(T)$ to within 1% accuracy.

TABLE I: Parameters for Potential Energy Barriers

system	$\Delta V,^a$ kcal/mol	$V_b,^b$ kcal/mol	$\alpha,^c$ a_0^{-1}	$\mu,^d$ amu	ref for system modeled
Eckart Approximations of CVE Potential Energy Barriers					
H + H ₂ , T + T ₂	0.0	9.80	2.10	0.6720, 2.011	32
Cl + H ₂ , Cl + T ₂	3.01	4.66	4.99	1.906, 5.145	33, 34
O + H ₂	-5.30	5.49	5.49	0.9512	35
O + HC	-9.20	6.26	5.00	7.034	35
Br + HC	15.00	3.39	1.98	10.43	35
F + H ₂ , F + T ₂	-31.75	1.06	1.92	1.822, 4.578	36, 37
I + H ₂	35.82	0.06	10.6	1.984	38
H + Cl ₂ , T + Cl ₂	-48.64	2.42	1.34	0.9935, 2.892	39, 40
Eckart Approximations to VAG Potential Energy Barriers					
H + H ₂	0	6.55	1.57	0.6720	32
Cl + H ₂	1.05	2.53	2.38	1.906	33, 34
H + Cl ₂	-45.20	2.37	1.32	0.9935	39, 40
Spline Fit to VAG Potential Energy Barrier					
Cl + H ₂	1.05	2.53		1.906	33, 34

^a Endoergicity. ^b Intrinsic barrier height. ^c Range parameter. ^d Reduced mass.

4. Test Cases

The question of how to approximate the effective potential energy barrier for a separable reaction coordinate has been discussed elsewhere.^{1,12,16,18,27-31} In the present article we are not concerned with that question, but rather with the rapid and accurate calculation of $\kappa(T)$ once a model has been chosen for $V(s)$. We illustrate the use of our procedures for 16 representative test cases. Fifteen of these involve Eckart potentials, which are reviewed in Appendix 1. The parameters of the Eckart potential and the reduced masses are given in Table I. These are chosen to mimic for typical reactions either (i) the potential energy along the minimum-energy reaction path or (ii) the ground-state, minimum-energy-path vibrationally adiabatic potential curve for collinear reaction. For either model i or ii the actual parameters are determined such that the barrier width is about the same as computed from full potential energy surfaces³²⁻⁴⁰ for the reactions. Model i for the effective potential energy barrier is called the conservation-of-vibrational-energy (CVE) model, and model ii is called the vibrationally adiabatic ground-state (VAG) model. These models are discussed more fully elsewhere.^{12,31} We also consider a spline fit to the VAG potential energy barrier for Cl + H₂ as obtained from the potential energy surface of Stern, Persky, and Klein³³ and Baer.³⁴ This is an interesting case because this potential, illustrated in Figure 8 of ref 41, has a somewhat irregular shape, including two important local maxima, each about 2.5 kcal/mol higher than the product energy, and a local minimum about 2.0 kcal/mol higher than the product energy. Thus, for example, when using the quadrature scheme with $M = 19$, four of the six highest energy points involve four classical turning points. This potential provides a stringent test of the present approximation scheme.

5. Results

To test the semiclassical approximation to the transmission coefficient, we compare the results of the semiclassical calculations to results obtained by Boltzmann averaging quantum mechanical transmission probabilities. For the Eckart potentials, the quantum mechanical transmission probability can be obtained analytically and is given by eq A7-A10 (see Appendix 1). By replacing $T^S(E)$ by the quantal $T(E)$ in eq 12 we obtain an approximate quantal transmission coefficient $\kappa^R(T)$ which still involves the reflection assumption of eq 10 and the assumption that the transmission probability is unity for $E > 2V_{\max} - E_0$. Accurate quantum mechanical trans-

mission coefficients $\kappa(T)$ are obtained by converged numerical integration of the actual quantum mechanical transmission probabilities in eq 2. The results are given in Table II. Accurate quantum mechanical transmission coefficients for the spline fitted VAG potential energy barrier for Cl + H₂ have been given elsewhere.^{31,41} The present results for this case are compared to the quantum mechanical results in Table III.

6. Discussion

We consider first the results for the Eckart potentials. These systems are arranged in Tables I and II in roughly increasing order of $|\Delta V|$ and roughly decreasing order of V_b . The Eckart potentials were chosen for convenience in testing our procedures, but the conclusions should apply equally well to other shapes of potentials where the quantal transmission probabilities cannot be obtained analytically. In all cases we obtain good results at 1000 K so the discussion will concentrate on lower temperatures.

The first three test cases are symmetric. The errors in the semiclassical procedure are 5-12% at 200 K and 3-10% at 300 K. For these highly quantal systems the failure of the separability approximation itself leads to greater errors than this³¹ so the accuracy is acceptable. Notice that $\kappa^R(T)$ agrees with $\kappa(T)$ within 1% so that the reflection assumption causes little error here. The VAG barrier is flatter and broader than the CVE one,¹² and this different shape accounts for the smaller $\kappa(T)$ and the greater accuracy of the semiclassical approximation.

We also considered VAG and CVE barriers for the nearly symmetric system Cl + H₂ and the conclusions are similar to those drawn for H + H₂.

For H + Cl₂ the VAG and CVE barriers are very similar to each other, and this can be expected in general for very asymmetric reactions. In general, a representative range of effective potential energy barriers for collinear and three-dimensional asymmetric systems can be obtained by considering a set of CVE barriers. Since our goal here is to test the accuracy of the semiclassical method for a representative sample of cases, we will consider Eckart fits only to CVE barriers in the rest of the cases.

The reactions O + H₂, O + HC, and Br + HC form an intermediate set with $|\Delta V|$ in the range 5-15 kcal/mol. For the first two cases the errors are 11-12% at 200 K and less than or equal to 10% at temperatures of 300 K or higher. Again the reflection approximation is only a minor component of the error. For the third case, the barrier is broader, the reduced mass greater, and transmission coefficient closer to unity. In this case the semiclassical

TABLE II: Transmission Coefficients for Eckart Potentials

system	T, K	κ	κ^R	κ^S
Symmetric Systems				
H + H ₂ (CVE)	200	65.1	65.1	57.2
	300	45.9	45.4	41.3
	600	1.44	1.41	1.36
	1000	1.15	1.13	1.11
T + T ₂ (CVE)	200	3.20	3.18	3.03
	300	1.61	1.60	1.56
	600	1.13	1.12	1.11
	1000	1.05	1.04	1.03
H + H ₂ (VAG)	200	3.54	3.51	3.25
	300	1.71	1.68	1.61
	600	1.15	1.13	1.12
	1000	1.06	1.05	1.04
Slightly Asymmetric Systems				
Cl + H ₂ (CVE)	200	36.8	36.7	29.4
	300	4.94	4.86	4.11
	600	1.54	1.48	1.39
	1000	1.19	1.15	1.13
Cl + T ₂ (CVE)	200	4.34	4.29	3.88
	300	1.87	1.84	1.74
	600	1.18	1.16	1.14
	1000	1.07	1.05	1.05
Cl + H ₂ (VAG)	200	1.56	1.52	1.46
	300	1.23	1.20	1.18
	600	1.06	1.05	1.04
	1000	1.03	1.02	1.01
O + H ₂ (CVE)	200	13.60	13.54	11.91
	300	2.91	2.86	2.63
	600	1.31	1.28	1.25
	1000	1.11	1.09	1.08
Systems with $ V_1 > V_0$				
O + HC (CVE)	200	9.91	9.86	8.86
	300	2.58	2.54	2.38
	600	1.27	1.25	1.22
	1000	1.10	1.08	1.07
Br + HC (CVE)	200	1.12	1.12	1.12
	300	1.05	1.05	1.05
	600	1.01	1.01	1.01
	1000	1.01	1.00	1.00
F + H ₂ (CVE)	200	1.27	1.26	1.25
	300	1.11	1.11	1.10
	600	1.02	1.03	1.02
	1000	1.01	1.01	1.01
F + T ₂ (CVE)	200	1.11	1.11	1.10
	300	1.04	1.05	1.04
	600	1.01	1.01	1.01
	1000	1.00	1.00	1.00
I + H ₂ (CVE)	200	0.72	1.005	1.010
	300	0.75	1.002	1.004
	600	0.83	1.001	1.001
	1000	0.88	1.000	1.000
H + Cl ₂ (CVE)	200	1.67	1.66	1.63
	300	1.25	1.25	1.24
	600	1.06	1.06	1.05
	1000	1.02	1.02	1.02
T + Cl ₂ (CVE)	200	1.20	1.20	1.19
	300	1.08	1.08	1.08
	600	1.02	1.02	1.02
	1000	1.01	1.01	1.01
H + Cl ₂ (VAG)	200	1.62	1.61	1.59
	300	1.24	1.23	1.22
	600	1.05	1.05	1.05
	1000	1.02	1.02	1.02

approximation causes negligible error.

For the very asymmetric reactions, the intrinsic barrier height is usually small. In such cases it is instructive to think of the scattering as occurring in the endoergic direction. Then it is clear that reflection by the large part of the barrier due to the endoergicity can become just as significant as tunneling through the intrinsic barrier. This makes $\kappa(T)$ generally smaller than for more symmetric cases. For asymmetric cases, the semiclassical approxi-

TABLE III: Transmission Coefficients for Spline Fit to the VAG Potential for Collinear Cl + H₂

T, K	κ	κ^S
200	1.69 ^a	1.49
300	1.32 ^a	1.19
600	1.11 ^a	1.04
1000	1.05 ^a	1.02

^a Reference 31.

mation can be quite accurate. For example, for F + H₂, F + T₂, H + Cl₂, and T + Cl₂, the errors are all less than 2%. However, when the intrinsic barrier gets very small our assumption for reflection probabilities becomes poor. For I + H₂ the intrinsic barrier is only 0.06 kcal/mol, and thus our reflection assumption corresponds to unit transmission probability at all energies 0.06 kcal/mol or more above the barrier height. This is a poor approximation. The semiclassical prediction that the transmission probability is 0.5 for an energy equal to the barrier height is also a poor approximation in a case like this. For these reasons the semiclassical approximation overestimates the thermally averaged transmission coefficient in this case. It should be noted, however, that the use of a transmission coefficient calculated on the basis of the separable approximation does not give good results in this case even when the transmission coefficient is calculated by accurate quantum mechanics.⁴¹

Table III shows the results for the spline fit to the VAG barrier for Cl + H₂. The errors are 3–12% which are comparable to the errors for the Eckart fits for other nearly symmetric or symmetric cases. This is especially encouraging since the spline fit has a double maximum for this case as discussed in section 5. Although the error is larger for the true potential than the Eckart fit for this case, it is still small enough to illustrate an important point. We have shown elsewhere³¹ that, even by using accurate quantum mechanics and the best available models for a transmission coefficient based on a separable reaction coordinate, the predicted rate constant for collinear reaction differs from the exact quantal rate for the same potential energy surface by a factor of 2.0 at 300 K. In the next paragraph we show that the execution time used to calculate the quantal transmission coefficient is about 200 times greater than that necessary for the semiclassical calculation. With this great disparity in cost, one will generally be willing to accept inaccuracies due to the semiclassical approximation whenever they are smaller than the other inaccuracies due to the transition state theory formalism.

Table IV illustrates the convergence properties of the present quadrature scheme. Convergence is slowest at low temperature so only 200 and 300 K are shown. In a given calculation we use $(2n + 1)$ -point quadrature for the transmission probability at each energy and $(2m + 1)$ -point quadrature for the thermal average, and we also print out the other available pairs of answers, viz., (n, m) , $(n, 2m + 1)$, and $(2n + 1, m)$. Comparing the various results shows that 1% convergence can be obtained with $n = m = 4$. For $n = m = 9$ all four points of results agree within 0.3% at both temperatures. For $n = m = 7$ (not shown), this increases to 0.7%, and for $n = m = 5$ and 4, it increases to 3 and 4%, respectively. Thus the error limits estimated by comparing the four answers obtained in one calculation are conservative and therefore reliable. The execution time required to obtain the four answers for $n = m = 7$ at eight temperatures is 0.27 s on the Control Data Corporation Cyber 74 computer. For $n = m = 9$ this increases to 0.37 s. This compares quite favorably to the computer time we used for the quantum mechanical calculations on the same

TABLE IV: Convergence of Transmission Coefficient with Respect to Quadrature Size for Spline Fit to the VAG Barrier for Collinear Cl + H₂

m^b	M^b	$n^a = 4$		$n = 5$		$n = 9$	
		$N^a = 4$	$N = 9$	$N = 5$	$N = 11$	$N = 9$	$N = 19$
$T = 200 \text{ K}$							
4	4	1.556	1.500	1.470	1.494	1.498	1.496
	9	1.526	1.493	1.472	1.487	1.491	1.489
5	5	1.585	1.476	1.438	1.464	1.471	1.470
	11	1.558	1.486	1.456	1.477	1.482	1.481
9	9	1.519	1.489	1.470	1.485	1.488	1.486
	19	1.532	1.487	1.465	1.482	1.485	1.483
$T = 300 \text{ K}$							
4	4	1.216	1.199	1.189	1.196	1.198	1.197
	9	1.203	1.193	1.187	1.191	1.192	1.192
5	5	1.225	1.186	1.173	1.182	1.184	1.184
	11	1.215	1.190	1.181	1.187	1.189	1.189
9	9	1.201	1.192	1.186	1.190	1.191	1.190
	19	1.205	1.191	1.184	1.189	1.190	1.190

^a Quadrature parameters for calculation of transmission probability, eq 4 and 22. ^b Quadrature parameters for calculation of transmission coefficients, eq 12.

problem. For that calculation we calculated accurate quantum mechanical transmission probabilities at 51 energies and Boltzmann averaged them numerically. This required about 80 s execution time on the Cyber 74 computer. The quantal calculations also require more personal attention to integration limits and other numerical parameters.

7. Concluding Remarks

We have shown that a semiclassical approximation provides a convenient and economical way to calculate thermally averaged transmission coefficients for separable potential energy barriers. The numerical error in performing the semiclassical calculations can easily be monitored and kept below 1% or less. The error caused by the semiclassical approximation as compared to quantum mechanics is generally 10% or less for typical potential energy barriers representing chemical reactions. The computer program KAPPAS for calculating semiclassical transmission coefficients by the method described here is being submitted to the National Resource for Computation in Chemistry (NRCC), Lawrence Berkeley Laboratory, Berkeley, Calif. 94720. A copy of the program and a test run may be obtained from them. The program KAPPAS will accept any potential function having only a single maximum, although it can readily be generalized as was the program used for the results reported here.

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Appendix 1

An Eckart potential has the form

$$V(s) = \frac{\Delta V y}{1 + y} + \frac{B y}{(1 + y)^2} \quad (\text{A1})$$

where

$$y = \exp[\alpha(s - s_0)] \quad (\text{A2})$$

$$B = [V_{\max}^{1/2} + (V_{\max} - \Delta V)^{1/2}]^2 \quad (\text{A3})$$

ΔV is the value of the potential at $s = +\infty$ relative to $s = -\infty$, $V_{\max}$ is the value of the potential at its maximum relative to $s = -\infty$, and the zero of energy is chosen to be the value of the potential at $s = -\infty$. The constant s_0 is arbitrary and we chose it to give the maximum of the

Eckart potential at $s = 0$, thus

$$s_0 = (2\alpha)^{-1} \ln [(V_{\max} - \Delta V)/V_{\max}] \quad (\text{A4})$$

The range of the potential is determined by α which is related to the magnitude F_s of the second derivative of the potential at its maximum by

$$\alpha^2 = BF_s/[2V_{\max}(V_{\max} - \Delta V)] \quad (\text{A5})$$

The intrinsic barrier height is

$$V_b = V_{\max} - \max [0, \Delta V] \quad (\text{A6})$$

The natural parameters to specify the potential are ΔV , $V_{\max}$ or V_b , and α or F_s .

The quantum mechanical probability of transmission for a particle of mass m and energy E through the Eckart potential is given by^{6,7}

$$T(E) = \frac{\cosh(a + b) - \cosh(a - b)}{\cosh(a + b) + \cosh(d)} \quad (\text{A7})$$

where

$$a = 2\pi(2mE)^{1/2}/\hbar\alpha \quad (\text{A8})$$

$$b = 2\pi[2m(E - \Delta V)]^{1/2}/\hbar\alpha \quad (\text{A9})$$

$$d = 2\pi[2mB - (\hbar\alpha/2)^2]^{1/2}/\hbar\alpha \quad (\text{A10})$$

Notice that d becomes purely imaginary for $2mB < \hbar^2\alpha^2/4$; then $\cosh d$ becomes $\cos |d|$ in eq A7.

For energies less than $V_{\max}$ it is possible to derive an analytic expression for the barrier penetration integral $\theta(E)$ and thus for the semiclassical transmission probability for the Eckart potential. The barrier penetration integral is given by

$$\theta(E) = \frac{1}{2}(f - b - a) \quad E < V_{\max} \quad (\text{A11})$$

where a and b are given in eq A8 and A9 and

$$f = 2\pi[2mB]^{1/2}/\hbar\alpha \quad (\text{A12})$$

Supplementary Material Available: Appendix 2 contains a listing of the Fortran program which is available from NRCC. The operation of the program is described on comment cards. The listing also includes a test potential, test data, and the output of a test run (18 pages). Ordering information for Appendix 2 is available on any current masthead page.

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Circular Dichroism of a Chiral Alkylbenzene. A Coupled Oscillator Approach

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The dynamic coupling of electric dipole allowed benzene transitions with those of the neighboring alkyl moiety is shown to be an important, possibly predominant, mechanism for optical activity in the (+)-(S)-2-phenyl-3,3-dimethylbutane compound by means of a polarizability treatment. A model of the alkyl polarizabilities to be used in the calculations is presented and discussed. Comparison of the results obtained is made both with circular dichroism and with absorption UV spectra determined experimentally.

Introduction

The (+)-(S)-2-phenyl-3,3-dimethylbutane compound (I), Figure 1, shows an intense circular dichroism (CD) in correspondence to the electrically allowed bands in the 50×10^3 – 54×10^3 -cm⁻¹ frequency region. The maximum value of $\Delta\epsilon$ is $24 \text{ L}\cdot\text{mol}^{-1}\cdot\text{cm}^{-1}$ at $52.2 \times 10^3 \text{ cm}^{-1}$, and the rotatory strength of the band is 16×10^{-40} (cgs).¹

The entire observed UV spectrum is typically aromatic; the frequency position, the dipolar strength, and the shape of the bands are very similar to those of the parent compound benzene. These similarities suggest an interpretation of the UV and CD spectra in terms of an independent-system model,^{2,3} i.e., in terms of an aromatic moiety interacting with a chiral aliphatic system.

The central problem is the representation of the aromatic and aliphatic subsystems and of their interaction mechanism.⁴ First-order perturbative quantum-mechanical computations,³ a truly independent system treatment, have already been carried out on benzene chirally perturbed chromophores such as (S)-(+)-1-methylindane^{5,6} and (+)-(2)-methylenbenzborbornene.⁷ It must be observed that in such molecules, as well as in I, the dimensions of the chromophores are of the same order of magnitude as the interchromophoric distances, a condition which is normally adverse to independent system approaches.⁴ This combination of contradictory geometric and spectroscopic features indicates then the possibility of an intermediate situation between the case of only one dissymmetric chromophore and that of two separate, interacting ones. So it seems expedient to preserve the independent system picture as suggested by the UV spectrum while treating to all orders the interaction between the subsystems. There are various all-order polarizability theories.^{8,9}

In the classical (nonquantum) coupled oscillators model developed by DeVoe,⁸ each electronic transition is represented by an electron oscillator whose polarizability may

be derived from the UV spectra of model compounds. These oscillators, distributed on the relevant chromophores, interact with each other to all orders when immersed in electromagnetic radiation. The DeVoe treatment has been shown to be suitable for treating the coupling of electric dipole allowed transitions localized on the different chromophores.¹⁰⁻¹³ Indeed, the dynamic coupling mechanism¹⁴ in the above-mentioned first-order computations^{5,7} is singled out as being the dominant source of CD in the region of strong aromatic absorption. So it is of interest to see if this polarizability approach is able to interpret the observed CD spectrum, at least in the region of strong aromatic absorption (50×10^3 – $54 \times 10^3 \text{ cm}^{-1}$).

Oscillator Polarizabilities

In the present development the aromatic absorption band at 48×10^3 – $54 \times 10^3 \text{ cm}^{-1}$ ($^1A_{1g} \rightarrow ^1E_{1u}$ in benzene) has been resolved into two components:¹⁰ $^1A_1 \rightarrow ^1B_1$ and $^1A_1 \rightarrow ^1A_1$ (C_{2v} local symmetry), both centered at the same frequency, $53 \times 10^3 \text{ cm}^{-1}$, but respectively with perpendicular and parallel polarization relative to the C_2 local axis. A third transition, centered at $49 \times 10^3 \text{ cm}^{-1}$ ($^1A_1 \rightarrow ^1A_1$ in character), is assumed to borrow its intensity from the above parallel and neighboring $^1A_1 \rightarrow ^1A_1$ transition.¹⁰ A further polarizability (orthogonal to the C_2 axis) has been considered in order to take into account also the absorption at $38 \times 10^3 \text{ cm}^{-1}$. All these dipoles have been located at the center of the phenyl ring and the toluene spectrum has been used all along to obtain the polarizability data.¹⁰

As far as the alkyl part is concerned, methane and 2-methylpropane have been assumed as suitable model chromophores for the methyl and *tert*-butyl groups, respectively. Their UV spectra are known up to $93 \times 10^3 \text{ cm}^{-1}$,¹⁵ but a detailed knowledge of the transitions involved and of the corresponding polarization directions is still lacking. In other words, with the models used to interpret